

Vel Tech Group of Institutions

Name: GOPICHAND KAKIVAI

Email: vtu29748@veltech.edu.in

Roll no: vtu29748

Phone: null

Branch: VTU

Department: CSE

Batch: 2028

Degree: B.Tech - Computer Science and Engineering

Scan to verify results



2028_NeoColab_Design and Analysis of AlgorithmsS1L3

VTU_DAA_Week 10_MCQ

Attempt : 1

Total Mark : 15

Marks Obtained : 14

Section 1 : MCQ

1. Which of the following methods is commonly used to solve assignment problems?

Answer

Hungarian method

Status : Correct

Marks : 1/1

2. The application of assignment problems is to obtain _____

Answer

Minimum cost or maximum profit

Status : Correct

Marks : 1/1

3. The assignment problem is said to be balanced if _____.

Answer

the number of rows is equal to the number of columns

Status : Correct

Marks : 1/1

4. The minimum number of lines covering all zeros in a reduced cost matrix of order n can be _____.

Answer

at the most n

Status : Correct

Marks : 1/1

5. The assignment problem _____.

Answer

All of the mentioned options

Status : Correct

Marks : 1/1

6. You are given a knapsack that can carry a maximum weight of 60. There are 4 items with weights of {20, 30, 40, and 70} and values {70, 80, 90, and 200}.

What is the maximum value of the items you can carry in the knapsack?

Answer

160

Status : Correct

Marks : 1/1

7. What is the objective of the knapsack problem?

Answer

To Get Maximum Total Value In The Knapsack

Status : Correct

Marks : 1/1

8. Which of the following methods is used to prune the search space in the Branch and Bound method for the Knapsack problem?

Answer

Ignoring nodes with a weight exceeding the capacity.

Status : Correct

Marks : 1/1

9. What is the key difference between Backtracking and Branch and Bound in solving the Knapsack problem?

Answer

Backtracking does not use any bounding function, while Branch and Bound does.

Status : Correct

Marks : 1/1

10. What is the primary advantage of using the Branch and Bound approach for the Knapsack problem?

Answer

It systematically explores possible solutions while pruning non-promising ones.

Status : Correct

Marks : 1/1

11. The travelling salesman problem can be solved using _____.

Answer

a minimum spanning tree

Status : Correct

Marks : 1/1

12. In a traveling salesman problem, the elements of diagonal from left-top to right-bottom are _____.

Answer

all infinity

Status : Correct

Marks : 1/1

13. Which type of graph is used to model the TSP problem?

Answer

Directed Weighted Graph

Status : Wrong

Marks : 0/1

14. What is the main goal of using Branch and Bound in TSP?

Answer

To minimize search space by pruning unpromising paths

Status : Correct

Marks : 1/1

15. Which value helps in deciding whether a node should be pruned in Branch and Bound?

Answer

Bound (lower bound of cost)

Status : Correct

Marks : 1/1